

Some Properties of the Harmonic Series

Introduction

The series $H(n) = H(1, n) = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ is a *harmonic series*. Or more generally $H(m, n) = \frac{1}{m} + \frac{1}{m+1} + \dots + \frac{1}{n}$ is also called a *harmonic series*, here we always assume $1 \leq m < n$. There are many interesting properties concerning this series.

Example 1: $H(1, n)$ is unbounded, i.e. for any positive number A , we can find n big enough, so that $H(1, n) \geq A$.

Indeed for any positive integer r , $\frac{1}{r+1} + \frac{1}{r+2} + \dots + \frac{1}{2r} > \frac{13}{24} > \frac{1}{2}$, which can be proved by induction. Hence we can take enough pieces of these fractions to make $H(1, n)$ as large as possible.

Example 2: $H(m, n)$ is never an integer.

(i) For the special case $m=1$, let s be such that $2^s \leq n < 2^{s+1}$. We then multiply $H(1, n)$ by $2^{s-1}Q$, where Q is the product of all odd factors of 1 to n . All terms in $H(1, n)$ will become an integer except the term 2^s will become an integer divided by 2 (a half integer). This implies $H(1, n)$ is not an integer.

(ii) Again assume $m=1$, Denote by p the greatest prime number not exceeding n . By Bertrand's postulate there is a prime q with $p < q < 2p$. Therefore we have $n < 2p$.

If $H(1, n)$ is an integer, then $n!H(n) = \sum_{i=1}^n \frac{n!}{i}$ is an integer divisible by p . However

the term $\frac{n!}{p}$ (an addend) is not divisible by p but all other addends are.

(iii) We deal with the general case. Indeed suppose $2^\alpha | k$ but $2^{\alpha+1}$ does not divide k

(write this as $2^\alpha \| k$), then α is the "parity order" of k . Now observe

$2^\alpha, 3 \cdot 2^\alpha, 5 \cdot 2^\alpha, \dots$ all have the same parity order. In between these numbers are $2 \cdot 2^\alpha, 4 \cdot 2^\alpha, 6 \cdot 2^\alpha, \dots$, all have greater parity orders. Hence between any two numbers of the same parity order, there is one with greater parity order. This implies there is one number with the greatest parity order, say q of parity order μ . Now multiply

$\frac{1}{m} + \frac{1}{m+1} + \dots + \frac{1}{n}$ by $2^\mu L$, where L is the product of all odd factors of numbers from m to n , we then get $2^\mu L \cdot H(m, n)$, which is an odd number, hence $H(m, n) = \frac{2r+1}{2^\mu L} = \frac{q}{p}$, where p is even and q is odd and so is not an integer.

Example 3 (APMO 1997)

$$\text{Given that } S = 1 + \frac{1}{1+\frac{1}{3}} + \frac{1}{1+\frac{1}{3}+\frac{1}{6}} + \dots + \frac{1}{1+\frac{1}{3}+\frac{1}{6}+\dots+\frac{1}{1993006}},$$

where the denominators contains partial sum of the sequence of reciprocals of triangular numbers. Prove that $S > 1001$.

Let T_n be the n th triangular number. Then $T_n = \frac{n(n+1)}{2}$ and hence

$$\begin{aligned} \frac{1}{T_1} + \frac{1}{T_2} + \dots + \frac{1}{T_n} &= \frac{2}{1 \cdot 2} + \frac{2}{2 \cdot 3} + \dots + \frac{2}{n(n+1)} \\ &= 2\left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1}\right) = 2\left(1 - \frac{1}{n+1}\right) = \frac{2n}{n+1}. \end{aligned}$$

Since $1993006 = \frac{1996 \cdot 1997}{2}$, get

$$S = \frac{1}{2} \left(\frac{2}{1} + \frac{3}{2} + \dots + \frac{1997}{1996} \right) = \frac{1}{2} \left(1996 + \left(1 + \frac{1}{2} + \dots + \frac{1}{1996}\right) \right).$$

Hence $S > \frac{1}{2}(1996 + 6) = 1001$, by using example 1 to see that $\sum_{k=2^l+1}^{2^{l+1}} \frac{1}{k} > \frac{1}{2}$.

Divisibility Properties

Congruence relations of harmonic series are of some interest. First an example:

Example 4 (IMO 1979)

Let p, q be natural numbers such that $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{1318} + \frac{1}{1319}$. Prove that p is divisible by 1979.

We will prove the well-known identity (N. Botez (1982) and later used by Catalan):

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}.$$

It is proved as follows.

$$\begin{aligned} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n} &= (1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2n}) - 2(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2n}) \\ &= (1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2n}) - (1 + \frac{1}{2} + \dots + \frac{1}{n}) \\ &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}. \end{aligned}$$

Thus

$$\begin{aligned} \frac{p}{q} &= \frac{1}{660} + \frac{1}{661} + \dots + \frac{1}{1318} + \frac{1}{1319} \\ &= \frac{1}{2} \left(\frac{1}{660} + \frac{1}{1319} + \frac{1}{661} + \frac{1}{1318} + \dots + \frac{1}{1319} + \frac{1}{660} \right) \\ &= \frac{1}{2} \left(\frac{1979}{660 \cdot 1319} + \frac{1979}{661 \cdot 1318} + \dots + \frac{1979}{1319 \cdot 660} \right) \\ &= 1979 \cdot \frac{A}{B}, \end{aligned}$$

where B is the product of positive integers less than 1979. But 1979 is prime, hence $1979 \mid p$.

Another proof using congruence relations: $(k, 1979) = 1$, Fermat's little theorem

implies $k^{1978} \equiv 1 \pmod{1979}$, hence $\frac{1}{k} \equiv k^{1977} \pmod{1979}$, then

$$\begin{aligned} \sum_{k=1}^{1319} (-1)^{k-1} \frac{1}{k} &\equiv \sum_{k=1}^{1319} (-1)^{k-1} k^{1977} = \sum_{k=1}^{1319} k^{1977} - 2 \sum_{k=1}^{659} (2k)^{1977} \\ &= \sum_{k=1}^{1319} k^{1977} - 2 \cdot 2^{1977} \sum_{k=660}^{1319} k^{1977} \equiv \sum_{k=1}^{1319} k^{1977} - \sum_{k=1}^{659} k^{1977} \\ &= \sum_{k=660}^{1319} k^{1977} = \sum_{k=660}^{989} (k^{1977} + (1979-k)^{1977}) \equiv \sum_{k=660}^{989} (k^{1977} + (-k)^{1977}) \\ &\equiv 0 \pmod{1979}. \end{aligned}$$

Note that $\frac{1}{k} \pmod{p}$ (as well as any fraction mod p) makes sense if $k \not\equiv 0 \pmod{p}$.

Also a generalization, we have

Example 5: $H(m, n) = \frac{q}{p}$ and $m+n$ an odd prime number, then $m+n \mid q$.

Note that $H(m, n)$ has an even number of terms and

$$\begin{aligned}
 H(m, n) &= \left(\frac{1}{m} + \frac{1}{n} \right) + \left(\frac{1}{m+1} + \frac{1}{n-1} \right) + \dots + \left(\frac{1}{\frac{m+n-1}{2}} + \frac{1}{\frac{m+n+1}{2}} \right) \\
 &= (m+n) \left(\frac{1}{mn} + \frac{1}{(m+1)(n-1)} + \dots + \frac{1}{\frac{m+n-1}{2} \cdot \frac{m+n+1}{2}} \right) \\
 &= (m+n) \frac{s}{r},
 \end{aligned}$$

where $\gcd(s, r) = 1$. Now note that $\gcd(r, m+n) = 1$, hence $\frac{q}{p} = \frac{(m+n)s}{r}$ and

$$m+n \mid q.$$

Now, The Catalan's Identity is used elsewhere.

Example 6 (Rom Math Magazine, July 1998):

$$\text{Let } A = \frac{1}{1 \cdot 2} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{2011 \cdot 2012} \text{ and}$$

$$B = \frac{1}{1007 \cdot 2012} + \frac{1}{1008 \cdot 2011} + \dots + \frac{1}{2012 \cdot 1007}.$$

Evaluate $\frac{A}{B}$.

$$\begin{aligned}
 A &= \frac{1}{1 \cdot 2} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{2011 \cdot 2012} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2011} - \frac{1}{2012} \\
 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2011} + \frac{1}{2012} - 2\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2012}\right) \\
 &= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2011} + \frac{1}{2012}\right) - \left(1 + \frac{1}{2} + \dots + \frac{1}{1006}\right) \\
 &= \frac{1}{1007} + \frac{1}{1008} + \dots + \frac{1}{2012} \quad (\text{Catalan's Identity}) \\
 &= \frac{1}{2} \left(\frac{1}{1007} + \frac{1}{2012} + \frac{1}{1008} + \frac{1}{2011} + \dots + \frac{1}{2012} + \frac{1}{1007} \right) \\
 &= \frac{1}{2} \left(\frac{3019}{1007 \cdot 2012} + \frac{3019}{1008 \cdot 2011} + \dots + \frac{3019}{2012 \cdot 1007} \right) \\
 &= \frac{3019B}{2}.
 \end{aligned}$$

Hence $\frac{A}{B} = \frac{3019}{2}$.

An important congruence relation concerning the harmonic series is the theorem of Wolstenholme:

Theorem (Wolstenholme): For $p \geq 5$, $H(p-1) = 1 + \frac{1}{2} + \dots + \frac{1}{p-1} \equiv 0 \pmod{p^2}$.

Or if $H(p-1) = 1 + \frac{1}{2} + \dots + \frac{1}{p-1} = \frac{a}{b}$, then $p^2 | a$.

(Examples:

$$H(4) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12} \text{ and } 5^2 | 25.$$

$$H(6) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{49}{20} \text{ and } 7^2 | 49.$$

$$H(10) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{10} = \frac{7381}{2520} \text{ and } 11^2 | 7381.)$$

(i) First proof:

$$H(p-1) = \sum_{n=1}^{p-1} \frac{1}{n} = \sum_{n=1}^{(p-1)/2} \left(\frac{1}{n} + \frac{1}{p-n} \right) = p \sum_{n=1}^{(p-1)/2} \frac{1}{n(p-n)}.$$

So we need to prove $\sum_{n=1}^{(p-1)/2} \frac{1}{n(p-n)} \equiv 0 \pmod{p}$.

$$\text{But } \sum_{n=1}^{(p-1)/2} \frac{1}{n(p-n)} \equiv - \sum_{n=1}^{(p-1)/2} \frac{1}{n^2} \pmod{p}.$$

$$\text{Now } \sum_{n=1}^{(p-1)/2} \frac{1}{n^2} \equiv \sum_{n=1}^{(p-1)/2} n^2 \equiv \frac{p(p-1)(2p-1)}{6} \equiv 0 \pmod{p}.$$

Note that the numbers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are all the quadratic residues and so are their reciprocals. And their sum is $0 \pmod{p}$.

(ii) Second proof, using polynomials mod p :

We need a theorem of Lagrange, if $f(x) = c_0 + c_1x + \dots + c_nx^n$ is a polynomial of degree n , with integer coefficients, and if the congruence $f(x) \equiv 0 \pmod{p}$ has more than n solutions, where p is prime, then every coefficient of $f(x)$ is divisible by p . The proof is not hard, it can be done basically by induction and the division algorithm

mod p . The statement is not true if p is not prime. For instance, $x^2 - 1 \equiv 0$ has 4 solutions mod 8. Now another proof of Wolstenholme:

From Fermat's Little theorem, $x^{p-1} \equiv 1 \pmod{p}$ has $1, 2, \dots, p-1$ as solutions.

Thus $x^{p-1} - 1 \equiv (x-1)(x-2)\dots(x-p+1) \pmod{p}$. Let

$$(x-1)(x-2)\dots(x-p+1) = x^{p-1} - s_1 x^{p-2} + \dots - s_{p-2} x + s_{p-1} \quad (*).$$

By Wilson's theorem $s_{p-1} = (p-1)! \equiv -1 \pmod{p}$. Thus

$$0 \equiv s_1 x^{p-2} - s_2 x^{p-3} + \dots - s_{p-3} x^2 + s_{p-2} x \pmod{p}.$$

The formula is true for all $x \in \mathbb{N}$, by Lagrange, p divides each of $s_1, s_2, \dots, s_{p-2}$.

Put $x = p$ in (*), get $(p-1)! = p^{p-1} - s_1 p^{p-2} + \dots - s_{p-2} p + s_{p-1}$.

Canceling out $(p-1)!$ and dividing both sides by p , get $0 = p^{p-2} - s_1 p^{p-3} + \dots - s_{p-2}$.

As $p \geq 5$, there are enough terms each of them congruent to 0 mod p^2 , hence get

$s_{p-2} \equiv 0 \pmod{p^2}$. And as

$$s_{p-2} = (p-1)! \left(1 + \frac{1}{2} + \dots + \frac{1}{p-1}\right) = (p-1)! \frac{a}{b}. \quad \text{This proves the theorem of}$$

Wolstenholme.

(Using Wolstenholme, and by putting $x = kp$ in (*), we get

$$\begin{aligned} \prod_{i=1}^{p-1} (kp - i) &= (kp-1)(kp-2)\dots(kp-p+1) \\ &= (kp)^{p-1} - s_1 (kp)^{p-2} + \dots + s_{p-3} (kp)^2 - s_{p-2} kp + s_{p-1} \\ &\equiv s_{p-3} k^2 p^2 - s_{p-2} kp + s_{p-1} \equiv (p-1)! \pmod{p^3}. \end{aligned}$$

Upon dividing by $(p-1)!$, we have $\binom{kp-1}{p-1} \equiv 1 \pmod{p^3}$, $k = 1, 2, \dots$. The result

may in fact be taken as the statement of the theorem of Wolstenholme.

A few further remarks, the Wolstenholme's theorem or congruence of harmonic series are related to Bernoulli numbers B_n . For instance, we have

$$\binom{kp-1}{p-1} \equiv 1 - \frac{1}{3}(k^2 - k)p^3 B_{p-3} \pmod{p^4} \quad (\text{Glaisher's congruence}). \quad \text{These numbers are}$$

related to the Fermat's Last Theorem. Now we know for any prime $p \geq 5$, $\binom{kp-1}{p-1} \equiv 1 \pmod{p^3}$. Are there primes satisfying $\binom{2p-1}{p-1} \equiv 1 \pmod{p^4}$? If so, they are called *Wolstenholme Primes*. (so far, we only know 16843 and 2124679 are such primes). Yet another direction, can one find composite numbers n such that $\binom{kn-1}{n-1} \equiv 1 \pmod{n^3}$? At this point, no! All these are very classical things.)

Example 7 (APMO 2006): Let $p \geq 5$ be a prime and let r be the number of ways of placing p checkers on a $p \times p$ checkerboard so that not all checkers are in the same row (but they may all be in the same column). Show that r is divisible by p^5 .

Note that $r = \binom{p^2}{p} - p$. Hence it suffices to show that

$$(p^2 - 1)(p^2 - 2) \dots (p^2 - (p - 1)) - (p - 1)! \equiv 0 \pmod{p^4}. \quad (1)$$

Now let

$$f(x) := (x - 1)(x - 2) \dots (x - (p - 1)) = x^{p-1} + s_1 x^{p-2} + \dots + s_{p-2} x + s_{p-1}. \quad (2)$$

Thus the first congruence relation is the same as $f(p^2) - (p - 1)! \equiv 0 \pmod{p^4}$.

Therefore it suffices to show that $s_{p-2} p^2 \equiv 0 \pmod{p^4}$ or $s_{p-2} \equiv 0 \pmod{p^2}$, which is exactly the theorem of Wolstenholme.

Example 8 (Putnam 1996): Let p be a prime number greater than 3 and $k = \left\lfloor \frac{2p}{3} \right\rfloor$,

show that $\binom{p}{1} + \binom{p}{2} + \dots + \binom{p}{k} \equiv 0 \pmod{p^2}$

(For example $\binom{7}{1} + \binom{7}{2} + \binom{7}{3} + \binom{7}{4} = 7 + 21 + 35 + 35 = 98 = 2 \times 49 \equiv 0 \pmod{7^2}$.)

Idea:

$$\binom{p}{i} = \frac{p(p-1) \dots (p-i+1)}{1 \cdot 2 \cdot \dots \cdot i}. \quad \text{This is a multiple of } p \text{ if } 1 \leq i \leq p-1,$$

$$\text{but } \frac{(-1) \dots (-(i-1))}{1 \cdot 2 \cdot \dots \cdot i} \equiv (-1)^{i-1} \frac{1}{i}.$$

Hence to prove the congruence it suffices to show the sum $1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{k-1} \frac{1}{k}$

is congruent mod p to 0. Now note that $-\frac{1}{2i} \equiv \frac{1}{2i} + \frac{1}{p-i} \pmod{p}$. Thus one shows that we can replace the sum by $1 + \frac{1}{2} + \dots + \frac{1}{p-1} \equiv 0 \pmod{p}$, which is implied by the Wolstenholme's congruence relation.

(More detailed proof)

Let $H(n) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ and let $P(n) = 1 - \frac{1}{2} + \dots + (-1)^{n-1} \frac{1}{n}$. Then the problem is reduced to showing that for any $p > 3$, p divides the numerator of $P\left(\left\lceil \frac{2p}{3} \right\rceil\right)$. First we note that p divides the numerator of $H(p-1)$ because

$$\begin{aligned} 2H(p-1) &= \left(1 + \frac{1}{p-1}\right) + \left(\frac{1}{2} + \frac{1}{p-2}\right) + \dots + \left(\frac{1}{p-1} + 1\right) \\ &= \frac{p}{p-1} + \frac{p}{2(p-2)} + \dots + \frac{p}{p-1} \equiv 0 \pmod{p}. \end{aligned}$$

Now we have two cases.

(I) $p = 3n + 1$, then $\left\lceil \frac{2p}{3} \right\rceil = 2n$. So we must show p divides the numerator of $P(2n)$. Now

$$\begin{aligned} H(3n) - P(2n) &= 2\left(1 + \frac{1}{2} + \dots + \frac{1}{2n}\right) + \left(\frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{3n}\right) \\ &= \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right) + \left(\frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{3n}\right) \\ &= \left(1 + \frac{1}{p-1}\right) + \left(2 + \frac{1}{p-2}\right) + \dots + \left(\frac{1}{n} + \frac{1}{p-n}\right) \\ &= \frac{p}{p-1} + \frac{p}{2(p-2)} + \dots + \frac{p}{n(p-n)}. \end{aligned}$$

Hence p divides the numerator of both $H(3n)$ and $H(3n) - P(2n)$, thus the numerator of $P(2n)$, and we are done.

(II) $p = 3n + 2$, then $\left\lceil \frac{2p}{3} \right\rceil = 2n + 1$. So we must show p divides the numerator of $P(2n + 1)$.

Now

$$\begin{aligned} H(3n+1) - P(2n+1) &= 2\left(1 + \frac{1}{2} + \dots + \frac{1}{2n}\right) + \left(\frac{1}{2n+2} + \frac{1}{2n+2} + \dots + \frac{1}{3n+1}\right) \end{aligned}$$

$$\begin{aligned}
&= (1 + \frac{1}{2} + \dots + \frac{1}{n}) + (\frac{1}{2n+2} + \frac{1}{2n+2} + \dots + \frac{1}{3n+1}) \\
&= (1 + \frac{1}{p-1}) + (2 + \frac{1}{p-2}) + \dots + (\frac{1}{n} + \frac{1}{p-n}) \\
&= \frac{p}{p-1} + \frac{p}{2(p-2)} + \dots + \frac{p}{n(p-n)}.
\end{aligned}$$

Hence again p divides the numerator of $H(3n+1) - P(2n+1)$, and hence $P(2n+1)$. We are done.

Example 9: Let $p \geq 5$ be a prime, show that if $1 + \frac{1}{2} + \dots + \frac{1}{p} = \frac{a}{b}$, then $p^4 | ap - b$.

First we note that the sum of quadratic residues modulo any p ($p \geq 5$) is congruent to 0 mod p . Indeed let $r_1, r_2, \dots, r_k$ be the quadratic residues and $S = r_1 + r_2 + \dots + r_k$. But the numbers $4r_1, 4r_2, \dots, 4r_k$ are also the quadratic residues. So $S \equiv 4S \pmod{p}$, get $S \equiv 0 \pmod{p}$.

Now we need: $p^2 \left| (p-1)! \left(1 + \frac{1}{2} + \dots + \frac{1}{p-1} \right) \right.$, which is exactly the theorem of

Wolstenholme.

Finally $1 + \frac{1}{2} + \dots + \frac{1}{p-1} = p^2 \frac{x}{y}$, where x, y are integers with y not divisible by p . So

$$\frac{a}{b} = p^2 \frac{x}{y} + \frac{1}{p} \Rightarrow \frac{ap - b}{bp} = \frac{p^2 x}{y} \Rightarrow ap - b = \frac{p^3 bx}{y}.$$

But $p \nmid b$ as it is the denominator

of the fraction $\frac{1 \cdot 2 \cdot \dots \cdot (p-1) + 1 \cdot 3 \cdot 4 \cdot \dots \cdot p + \dots + 2 \cdot 3 \cdot 4 \cdot \dots \cdot p}{1 \cdot 2 \cdot \dots \cdot p}$. Hence we are done.

Example 10: Let p be an odd prime, $1 - \frac{1}{2^2} + \frac{1}{3^2} - \dots + (-1)^{p-2} \frac{1}{(p-1)^2} \equiv 0 \pmod{p}$.

(Indeed $1 - \frac{1}{4} = \frac{3}{4}$ for $p = 3$

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} = \frac{115}{144} \text{ for } p = 5$$

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36} = \frac{973}{1200} \text{ for } p = 7.)$$

The proof is not hard, Indeed:

$$\begin{aligned}
\sum_{k=1}^{p-1} (-1)^{k-1} \frac{1}{k^2} &= - \sum_{k=1}^{(p-1)/2} \left((-1)^k \frac{1}{k^2} + (-1)^{p-k} \frac{1}{(p-k)^2} \right) \\
&= - \sum_{k=1}^{(p-1)/2} (-1)^k \left(\frac{1}{k^2} - \frac{1}{(p-k)^2} \right) = - \sum_{k=1}^{(p-1)/2} (-1)^k \left(\frac{1}{k^2} - \frac{1}{k^2} \right) \equiv 0 \pmod{p}.
\end{aligned}$$

Estimates

Example 11: $\frac{2(n-m+1)}{m+n} < H(m, n) < \frac{m+n}{2mn}(n-m+1).$

Note that $H(m, n) = \frac{1}{2} \sum_{k=0}^{n-m} \left(\frac{1}{m+k} + \frac{1}{n-k} \right) = \frac{m+n}{2} \sum_{k=0}^{n-m} \frac{1}{(m+k)(n-k)}.$

But $(m+k)(n-k) \leq \left(\frac{m+k+n-k}{2} \right)^2 = \left(\frac{m+n}{2} \right)^2.$

Also $k(n-m-k) \geq 0$ implies $(n-k)(m+k) \geq mn$. Together we have

$$\frac{4(n-m+1)}{(m+n)^2} \cdot \frac{m+n}{2} \leq \frac{m+n}{2} \sum_{k=0}^{n-m} \frac{1}{(m+k)(n-k)} \leq \frac{m+n}{2} \cdot \frac{n-m+1}{mn}, \text{ simplify to get}$$

$$\frac{2(n-m+1)}{m+n} \leq \frac{m+n}{2} \sum_{k=0}^{n-m} \frac{1}{(m+k)(n-k)} = H(m, n) \leq \frac{(m+n)(n-m+1)}{2mn}. \text{ (Note that the inequalities are indeed strict.)}$$

(Take $m=1$, get $\frac{2n}{1+n} \leq H(1, n) \leq \frac{n(n+1)}{2n} = \frac{n+1}{2}.$)

Example 12 (Putnam): $n(n+1)^{\frac{1}{n}} - n < H(1, n) < n - (n-1)n^{-\frac{1}{n-1}}.$

Note $\frac{n+H(1, n)}{n} = \frac{(1+1) + (1+\frac{1}{2}) + \dots + (1+\frac{1}{n})}{n}$

$$> \sqrt[n]{(1+1)(1+\frac{1}{2}) \dots (1+\frac{1}{n})} = \sqrt[n]{2 \cdot \frac{3}{2} \cdot \dots \cdot \frac{n+1}{n}} = (n+1)^{\frac{1}{n}}, \text{ hence}$$

$$n+H(1, n) > n(n+1)^{\frac{1}{n}}, \text{ or } n(n+1)^{\frac{1}{n}} - n < H(1, n).$$

On the other hand,

$$\frac{n-H(1,n)}{n-1} = \frac{(1-\frac{1}{2})+(1-\frac{1}{3})+\dots+(1-\frac{1}{n-1})}{n-1}$$

$$> \sqrt[n-1]{(1-\frac{1}{2})(1-\frac{1}{3})\dots(1-\frac{1}{n-1})} = n^{-\frac{1}{n-1}},$$

hence $H(1,n) < n - (n-1)n^{-\frac{1}{n-1}}$.

Partitions

Example 13: Given any proper fraction $\frac{m}{n}$, where m, n are positive integers

satisfying $0 < m < n$, then it is the sum of fractions of the form $\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k}$,

where $x_1, x_2, \dots, x_k$ are distinct positive integers.

We use the “greedy method”. Indeed $\frac{1}{x_1} \leq \frac{m}{n} < \frac{1}{x_1-1}$, where x_1 is a positive integer.

If $\frac{1}{x_1} = \frac{m}{n}$, then the problem is done. Otherwise $\frac{m}{n} - \frac{1}{x_1} = \frac{mx_1 - n}{nx_1} = \frac{m_1}{nx_1}$, where

$m_1 = mx_1 - n < m$ and obviously $nx_1 > n$. Let x_2 be another positive integer such

that $\frac{1}{x_2} \leq \frac{m_1}{nx_1} < \frac{1}{x_2-1}$. The procedure can be repeated until

$m > m_1 > m_2 > \dots > m_k > 0$ and $\frac{m}{n} = \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_k}$, where $1 \leq k \leq m$.

(Note: writing $\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}$, we observe actually there are infinitely many

ways of writing any proper fractions as sum of fractions of this kind. These fractions are actually called *unit fractions* or *Egyptian fractions*.)

(Example: $\frac{1}{2} < \frac{5}{7} < 1$, and $\frac{5}{7} - \frac{1}{2} = \frac{3}{14}$. Then $\frac{1}{5} < \frac{3}{14} < \frac{1}{4}$, and $\frac{3}{14} - \frac{1}{5} = \frac{1}{70}$.

Hence $\frac{5}{7} = \frac{1}{2} + \frac{1}{5} + \frac{1}{70}$.)

Example 14: Remove those terms in $1 + \frac{1}{2} + \dots + \frac{1}{n} + \dots$ such that its denominator in

decimal expansion contains the digit “9”, then the sequence is bounded.

The integers within the interval $[10^{m-1}, 10^m - 1]$ are m -digit numbers. The first digit from the left cannot be the digits “0” and “9”, (8 choices), the other digits cannot contain “9”, hence nine choices 0, 1, 2, 3, 4, 5, 6, 7 and 8. Altogether there are $8 \cdot 9^{m-1}$ such

integers. The sum of their reciprocals is less than $\frac{8 \cdot 9^{m-1}}{10^{m-1}} = 8 \left(\frac{9}{10} \right)^{m-1}$. The sum of

reciprocals of all such numbers is therefore less than $\sum_{m=0}^{\infty} 8 \left(\frac{9}{10} \right)^m = \frac{8}{1 - \frac{9}{10}} = 80$.

Example 15: Let $m > 1$ be a positive integer. Show that $\frac{1}{m}$ is the sum of

consecutive terms in the sequence $\sum_{j=1}^{\infty} \frac{1}{j(j+1)}$.

Indeed $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}$, hence the problem is reduced to finding integers a and b

such that $\frac{1}{m} = \frac{1}{a} - \frac{1}{b}$ (*). One obvious solution is $a = m-1$ and $b = m(m-1)$. To

find other solutions of (*), we note that $\frac{1}{a} > \frac{1}{m}$, hence $m > a$. Let $a = m - c$, then

$b = \frac{m^2}{c} - m$. For each c satisfying $c \mid m^2$ and $1 \leq c \leq m$, there exists one and only

one pair of a and b satisfying (*), and because $a < b$, the representation is unique.

Now consider all factors of m^2 except m , there are $d(m^2) - 1$ of them. If c is one of them, then exactly one of c or $\frac{m^2}{c}$ will be less than m . Hence the number of

solutions of (*) is $\frac{d(m^2) - 1}{2}$. ($d(n)$ counts the number of factors of n .)

Exercises

1. If $p > 2$ a prime, a an integer such that $0 < a < p-1$, then

$$\binom{p-1}{a} \equiv (-1)^a \pmod{p}.$$

2. (Pretoria Camp July 2007) Given a prime number $p > 3$, show that p^2 divides

$$\sum_{k=1}^{p-1} k^{2p+1}.$$

3. (Can MO 2004) Given a prime number $p > 3$, then $\sum_{k=1}^{p-1} k^{2p-1} \equiv \frac{p(p+1)}{2} \pmod{p^2}$.

4. (Bul. Math Competition 2001) For any positive integer n the sum $1 + \frac{1}{2} + \dots + \frac{1}{n}$ is written in the form $\frac{p_n}{q_n}$, where p_n and q_n are relatively prime. (a) Show that p_{67} is not divisible by 3. (a) Find all n , for which p_n is divisible by 3.

5. (South Africa Problem Set 2003) Let $1 + \frac{1}{2} + \dots + \frac{1}{100} = \frac{a}{b}$, where $\gcd(a, b) = 1$.

Show that ab is not divisible by 5.

6. Prove that for an infinity of natural numbers n the numerator of the fraction

$1 + \frac{1}{2} + \dots + \frac{1}{n}$ has at least two prime factors.

7. Let $p \geq 5$ be a prime number, and $H(1, p) = \frac{s}{pr}$, show that $p^3 \mid r - s$.

8. (a) Let $p > 2$ be a prime, show that $\frac{2^p - 2}{p} \equiv 1 - \frac{1}{2} + \dots - \frac{1}{p-1} \pmod{p}$.

(b) Suppose $2^p - 2 \equiv 0 \pmod{p^2}$, and $1 - \frac{1}{2} + \dots - \frac{1}{p-1} = \frac{a}{b}$, show that $p \mid a$.

9. Show that $(n - m + 1) \left(\left(\frac{n+1}{m} \right)^{\frac{1}{n-m+1}} - 1 \right) < H(m, n) < (n - m + 1) \left(1 - \left(\frac{m-1}{n} \right)^{\frac{1}{n-m+1}} \right)$.

10. Show that $\frac{1}{2} < H(1, n) + H(1, m) - H(1, mn) < 1$.

11. (40th Putnam) Let $p_i \in [0, 1], i = 1, 2, \dots, n$, show that there exists $0 \leq x \leq 1$,

satisfying $\sum_{i=1}^n \frac{1}{|x - p_i|} \leq 8n(1 + \frac{1}{3} + \dots + \frac{1}{2n-1})$.

12. Show that the ten digits 0, 1, 2, ..., 9 appear infinitely often among the decimal expansions of all the prime numbers p . (Note: $\sum_p \frac{1}{p}$ diverges.)